

Referee #1 (Remarks to the Author):

This paper presents an impressive result in time keeping and Time and Frequency metrology, the direct observation in a single atomic clock of the gravitational redshift on a millimeter scale atomic sample. Its importance does not come from the redshift test per se at the 20% level but it represents a real breakthrough on the way to probe relativity and quantum mechanics over extended atomic wavefunctions.

This is enabled by a set of technical improvements over previous results achieved by the same group on a strontium optical lattice clock. Using an ultra-stable laser with several tens of seconds coherence time in the optical domain, a low power optical lattice, a large atom number sample, spatially resolved detection of the transition probability, and synchronous measurements, the frequency sensitivity reaches the impressive and previously unexplored range of $3.1 \cdot 10^{-18}$ @ 1 second and $7.6 \cdot 10^{-21}$ after averaging. This enables the authors to exhibit the gravitational gradient across the 1 mm size atomic sample. The differential frequency sensitivity represents a one-order of magnitude gain over previous work. The state superposition coherence exceeds 30 seconds and the presented atomic line Q-factor of $3.6 \cdot 10^{16}$ is the highest ever reached in any system. These advances are outstanding.

The paper is very well written; references are adequate and the Methods section gives details on the various shifts and data analysis. I have two major comments and small remarks.

The first major comment comes from the analysis of figure 3a. I understand that the gold dashed line is the calculated second order Zeeman shift gradient correction based on the formula given in the method section where the magnetic field is measured from the splitting of the optical $\pm$ m opposite spin Zeeman transitions.

From the graph, I see that the dashed line is very close to the fitted black line that is claimed to be the redshift gradient. In other words, it seems to me that the second-order Zeeman shift gradient has the same order of magnitude as the redshift gradient itself. Is this correct? Then how come that in the table on page 20 giving the list of systematics the second-order Zeeman shift gradient appears only as an uncertainty? In this respect, it would be useful to give the value of B and of the splitting in Hz between opposite spin states as well as the residual B field gradient. One would then have a sense of the actual magnetic field gradient along z and of the actual second-order Zeeman correction. Note that the dimension of χ on page 19 cannot be Hz but must have dimension Hz^{-1} as it is multiplied by Hz^2 in the formula. Please explain better how the gravitational shift is disentangled from the second-order Zeeman correction.

The second major comment deals with the sentence:

With a decade of advancements, today's leading clocks are poised to enable geodetic surveys of elevation at the sub-centimeter level on Earth, a result unobtainable with other current techniques²⁶.

This sentence is very strong and may not be correct. The satellite surveys leading to the EGM Earth models (CHAMP GRACE, GOCE) and averaged over spatial grids do reach sub cm potential difference resolution in non-mountainous regions. The advantage of using clocks is that the survey can be local-point to local-point at the sub-cm level potential difference with no spatial average. Please clarify.

Minor remarks:

In Fig 2 a, I have wondered if the linear slope were a direct imaging of the gravitational redshift but it is simply an example of a residual systematic gradient. Please be more specific in the caption.

caption of fig 3. The predicted redshift gradient is a full red line on my version of the manuscript and not dashed red line. On figure 3b what is the chi square per degree of freedom of this data ? Visually the one sigma error bar seems a little narrow to my taste.

Table in the method section, systematic budget. Why is the first line called statistical? I believe that the first line is rather the grav redshift.

In summary, I recommend publication of the paper in Nature after clarification of the points raised above.

Referee #2 (Remarks to the Author):

I am very pleased to review the manuscript where the gravitational redshift at the submillimeter scale is resolved for the first time. It is indeed a breakthrough not only for the optical clock community, but for much broader audience capturing imagination by the power of precision measurements in modern physics. The expected frequency difference is tiny (at 10^{-19} level) and it is indeed impressive, that one can not only measure the frequency gradients on the size of atomic cloud (this is demonstrated previously), but rectify the gravitational shift by a tight control on all other spatial frequency gradients. The manuscript is written clearly, all key experimental steps are professionally highlighted to involve even non-specialists. As a good sign, it is easy to follow the manuscript's logic chain and the main messages only from the figures without deep immersion in the text. The manuscript is definitely important for the physical community and I highly recommend it for publishing in Nature journal.

However, there are several questions and suggestions which may be taken into account.

1. In the previous work (for example [5]) authors used the term "differential frequency", while in the current manuscript the term "fractional frequency" is widely used (in the abstract and further in text). It sounds that the fractional frequency means $(f_1 - f_2)/f_1$, but further in the text the term "differential fractional frequency" appears which puzzles the reader.
 2. Page 3: It is mentioned that temperature of the atoms is ~ 100 nK. However in the supplementary (Atomic sample preparation) it is 800 nK. There is a significant difference.
 3. Page 3, the first paragraph: to my opinion the term "single clock stability" is incorrect in this context. It maybe it is "relative frequency instability".
 4. Page 3, the second paragraph: "We then adiabatically ramp the lattice to an operational depth of 12 Erec and prepare atoms into a single nuclear spin." In supplementary the sequence is different (first polarization and then ramping lattice depth down).
 5. Page 3, the second paragraph: at the end there is a link to a Fig.1a. However I do not see how Fig.1a is related to 6 μ m resolution and 15 lattice sites.
 6. Page 4, the first paragraph: I do not see how delocalization measurement and data from Fig. 1d is related to the main topic of the paper (at least without reading ref. 31, where, as expected, these measurements are done more rigorously and are used to characterize cancelling of two types of shifts related to collisions).
 7. Page 4, second paragraph: Again, how does this part is related to the gravitational shift measurements? Results from these measurements are not used further. If it is important it should be explained more clearly. It is also only 4 seconds more than one demonstrated in [15], so there is no significant improvements. This part in the current form seems to be not much important for the main topic discussion.
 8. Page 6, first paragraph: From a caption of Fig.3, one can understand that data in Fig.3b is not a raw gradient (as seems from the text), but after subtracting known systematics.
 9. Fig. 2c: What is the reason that the last point drops out? It looks like the data would be fitted better if a non-zero offset is introduced.
- I would also suggest to move Fig 4a closer to Fig 2c.
10. Fig 3a: It is hard to see if purple points are already shifted from green by the yellow line – it is advisable to make the plot somehow more readable. For example, one could make figure a wider by shrinking figure b.
- One can also see that the number of points at some regions (for example around 0.15 mm) are all below the fit, which looks like a frequency shift (not a statistical fluctuation). Is it reasonable to suggest some explanation of such features?
11. Page 6, second paragraph: "frequency gradient of $-1.30(18) \times 10^{-19}/\text{mm}$ " - this value is 1σ from that of binned analysis - why? Was it a different measurement?
 12. Fig. 4 It would be interesting to see the QPN limit
 13. Methods, in-vacuum cavity: If finesse is ~ 1300 than I would expect a power build up factor at least π times smaller. Does value " ~ 700 " refer to power of incident beam, or (a more conventional) comparison to a standing wave with only a back reflecting mirror? Anyway I do not see how it matches.

Further it is mentioned that cavity mirrors are AR coated @698 nm and that one of the mirrors is

rigidly mounted for phase reference for the clock laser. I do not understand how these two facts correspond.

The reference 36 is missing a journal name, and the reference 35 has no information about mirror shielding. And it would be helpful to mention what shielding parameters are used.

14. Analysis: it is not clear why data with excitation fraction above or below some values should be removed instead of being used for correction.

It would be interesting to have a look at the Gaussian fit of atomic profile at fig 1a, since the data seem to form a plateau in the center. Does cloud position fluctuate at shorter time scale?

15. Imaging, first paragraph: does this correspond to geometric calculations (lens magnification and real size of a camera pixel)? Some magnetic field gradient might change the fall acceleration.

16. DC Stark: means DC Stark shift gradient?

17. Density shift: it seems the minus sign is lost in the last equation.

Author Rebuttals to Initial Comments:

Referee 1

Comment 1

The first major comment comes from the analysis of figure 3a. I understand that the gold dashed line is the calculated second order Zeeman shift gradient correction based on the formula given in the method section where the magnetic field is measured from the splitting of the optical $\pm m$ opposite spin Zeeman transitions. From the graph, I see that the dashed line is very close to the fitted black line that is claimed to be the redshift gradient. In other words, it seems to me that the second-order Zeeman shift gradient has the same order of magnitude as the redshift gradient itself. Is this correct? Then how come that in the table on page 20 giving the list of systematics the second-order Zeeman shift gradient appears only as an uncertainty? In this respect, it would be useful to give the value of B and of the splitting in Hz between opposite spin states as well as the residual B field gradient. One would then have a sense of the actual magnetic field gradient along z and of the actual second-order Zeeman correction. Note that the dimension of χ_{SI} on page 19 cannot be Hz but must have dimension Hz^{-1} as it is multiplied by Hz^2 in the formula. Please explain better how the gravitational shift is disentangled from the second-order Zeeman correction.

The referee is correct that there was a typo in the 2nd order Zeeman coefficient. We have corrected the typo (Hz to Hz^{-1}) in the text.

We appreciate the feedback that 2nd order Zeeman corrections were not presented in the most transparent way. As the magnetic field may slowly fluctuate during the course of hour-long measurements, we apply 2nd order Zeeman corrections directly to every clock lock cycle in the same manner as the density shift corrections. The referee is correct that the 2nd order Zeeman corrections have a similar gradient as the redshift ($-7.0 \times 10^{-20}/\text{mm}$ on 8/13). This arises from a splitting between opposite m_F states of 12.7 mHz/mm (0.291 mG/mm). We have added a second paragraph in the 'Zeeman Shifts' section of the methods to provide the relevant quantitative information the referee has requested.

Comment 2

“With a decade of advancements, today’s leading clocks are poised to enable geodetic surveys of elevation at the sub-centimeter level on Earth, a result unobtainable with other current techniques²⁶.”

This sentence is very strong and may not be correct. The satellite surveys leading to the EGM Earth models (CHAMP GRACE, GOCE) and averaged over spatial grids do reach sub cm potential difference resolution in non-mountainous regions. The advantage of using clocks is that the survey can be local-point to local-point at the sub-cm level potential difference with no spatial average. Please clarify.

We thank the referee for bringing these important space-based measurements to our attention. We have amended the text to emphasize the complementarity of the local vs spatial averaged approaches:

“With a decade of advancements, today’s leading clocks are poised to enable local geodetic surveys of elevation at the sub-centimeter level on Earth, complimenting spatial averaging techniques²⁶.”

Reference 26 now cites a book review of these missions.

Comment 3

In Fig 2 a, I have wondered if the linear slope were a direct imaging of the gravitational redshift but it is simply an example of a residual systematic gradient. Please be more specific in the caption.

The Ramsey spectroscopy used for Fig. 2 employs a single spin state, precluding the rejection of vector shifts. Density shift corrections are not applied either. The background magnetic field gradient leads to the slope in excitation fraction. These details are indeed described in the methods section, hence with the consideration of space limit we respectfully decline to add additional text to the figure caption.

Comment 4

caption of fig 3. The predicted redshift gradient is a full red line on my version of the manuscript and not dashed red line. On figure 3b what is the chi square per degree of freedom of this data ? Visually the one sigma error bar seems a little narrow to my taste.

The reduced chi squared statistic is 2.99. Inflating the uncertainty by the reduced chi squared statistic would lead to 2.0e-20 uncertainty, less than the combined uncertainty in Table 1. We thus do not report the gradient measurement of Fig. 3b with inflated uncertainty.

The text of Fig. 3 has been updated to report this reduced chi square value, as well as an additional set of dotted lines to indicate total uncertainty from Table 1.

Comment 5

Table in the method section, systematic budget. Why is the first line called statistical? I believe that the first line is rather the grav redshift.

We initially called it statistical to reflect the fact that it represented statistical measurements without systematic corrections. To be more consistent, we have now changed 'statistical' to 'gradient' to reflect that this is from the processed data of Fig. 3. This connects clearly to the later entry 'corrected gradient' where the additional corrections in the table are applied, which is then compared to the known redshift.

Referee 2

Comment 1

In the previous work (for example [5]) authors used the term “differential frequency”, while in the current manuscript the term “fractional frequency” is widely used (in the abstract and further in text). It sounds that the fractional frequency means $(f_1-f_2)/f_1$, but further in the text the term “differential fractional frequency” appears which puzzles the reader.

We choose to emphasize the term fractional frequency to convey a dimensionless figure of merit for clock performance, in line with the recent measurements of reference 19. It is true that often a differential frequency is measured, and when it is normalized by the carrier frequency, this quantity becomes a differential fractional frequency. We have adjusted the caption of Figure 4 to use the phrase 'fractional frequency instability' to be consistent.

Comment 2

Page 3: It is mentioned that temperature of the atoms is ~100 nK. However in the supplementary (Atomic sample preparation) it is 800 nK. There is a significant difference.

The 800 nK temperature reported in the methods is the temperature for atoms initially loaded in the deep $300 E_{\text{rec}}$ trap for initial cooling. For clock spectroscopy, the lattice depth is adiabatically ramped down to $12 E_{\text{rec}}$ where the 100 nK temperature is measured and reported.

Comment 3

Page 3, the first paragraph: to my opinion the term “single clock stability” is incorrect in this context. It maybe it is “relative frequency instability”.

The reviewer is correct that single clock stability is too strong of a phrase since this stability was inferred from the comparison between two separate regions with our atomic ensemble, but has not been demonstrated between independent clocks. We have amended the language to “expected single clock stability” to reflect this.

Comment 4

Page 3, the second paragraph: “We then adiabatically ramp the lattice to an operational depth of $12 E_{\text{rec}}$ and prepare atoms into a single nuclear spin.” In supplementary the sequence is different (first polarization and then ramping lattice depth down).

We thank the referee for pointing out this potential confusion – the text is amended to reflect the fact that cooling and polarization were performed at the same time.

“Atoms are simultaneously cooled and polarized into a single nuclear spin before the lattice is adiabatically ramped to an operational depth of $12 E_{\text{rec}}$.”

Comment 5

Page 3, the second paragraph: at the end there is a link to a Fig. 1a. However I do not see how Fig. 1a is related to 6 μm resolution and 15 lattice sites.

We agree and have removed the reference to Fig. 1a.

Comment 6

Page 4, the first paragraph: I do not see how delocalization measurement and data from Fig. 1d is related to the main topic of the paper (at least without reading ref. 31, where, as expected, these measurements are done more rigorously and are used to characterize cancelling of two types of shifts related to collisions).

Fig. 1d actually serves an important purpose, to demonstrate that atoms loaded into this very shallow optical lattice remain coherent over a long time scale. It is important to show this for both the carrier and Wannier-Stark sidebands as only when the system is coherent and precisely controllable can we implement the cancellation of the density-related frequency shift by engineering the s-wave (off-site) and p-wave (on-site) interactions. Because of the space limit, we cannot report in this manuscript all of the measurements we have made on the density cancellation study. At the present time we are writing an independent manuscript to report a systematic study of this topic.

Comment 7

Page 4, second paragraph: Again, how does this part is related to the gravitational shift measurements? Results from these measurements are not used further. If it is important it should be explained more clearly. It is also only 4 seconds more than one demonstrated in [15], so there is no significant improvements. This part in the current form seems to be not much important for the main topic discussion.

There is a key difference between the work reported here and that of Ref. [15]. The coherence reported in Ref. [15] concerns a pair of atoms or at most a couple of pairs of atoms, while this paper describes a many-body system involving 100,000 atoms. It is a

breakthrough to have such a large number of atoms and yet still enjoy this record long coherence – it very clearly points to the future direction for further enhancement of this clock system. In some sense, this is perhaps the most exciting prospect of this work, that we still have room for improvement in the near future!

Also to be rigorous, Ref. [15] reported a coherence time of 19.5 s, and used modeling/post processing to suggest that they could extrapolate this value to a predicted single particle coherence time. We think the apples-to-apples comparison is 19.5 s to 37 s.

Comment 8

Page 6, first paragraph: From a caption of Fig.3, one can understand that data in Fig.3b is not a raw gradient (as seems from the text), but after subtracting known systematics.

We have amended the main text to clarify this point.

“For each we fit a linear slope and offset after processing for density shift and 2nd order Zeeman corrections, reporting the slope in Fig. 3b.”

Comment 9

Fig. 2c: What is the reason that the last point drops out? It looks like the data would be fitted better if a non-zero offset is introduced.

I would also suggest to move Fig 4a closer to Fig 2c.

We agree with the referee that the last point indicates a possible extension of atom coherence times. Notably this measurement is complicated by loss of atom numbers at long Ramsey dark times, modifying potential atomic interaction induced decoherence. A future study of such mechanism and development of an atomic coherence decay model with additional dephasing mechanisms is an exciting direction. For this paper we have chosen to use a simple exponential decay as demonstrated in previous work, which may underreport our full atomic coherence times. But until we perform a more extensive study, we have decided against reporting an aggressively long coherence time.

Comment 10

Fig 3a: It is hard to see if purple points are already shifted from green by the yellow line – it is advisable to make the plot somehow more readable. For example, one could make figure a wider by shrinking figure b.

One can also see that the number of points at some regions (for example around 0.15 mm) are all below the fit, which looks like a frequency shift (not a statistical fluctuation). Is it reasonable to suggest some explanation of such features?

We have taken the referee’s suggestion to increase the relative size of 3a, while also slightly decreasing the width of the 2nd Order Zeeman corrections line.

The referee correctly notes that some regions appear to have offsets from the fitted slope. Notably this is a single data set with nearly all data points within $\pm 2.5 \times 10^{-19}$, such

deviations are consistent with reported statistical error bars. Speculatively, at this level, small imperfections in imaging, cooling, clock excitation, and a variety of other sources may possibly fluctuate. We have included evaluations for additional uncertainties in Table 1 but choose not to speculate in the paper as to the source of possible variations that visually appear in the data. We note that this will be an important undertaking for more precise evaluations in the future as we would like to continue to push the measurement uncertainty towards 1×10^{-21} .

Comment 11

Page 6, second paragraph: "frequency gradient of $-1.30(18) \times 10^{-19}/\text{mm}$ " - this value is 1σ from that of binned analysis - why? Was it a different measurement?

The two-region analysis uses the same data as the binned analysis except for the 8/13 datapoint owing to the use of a slightly different duty cycle.

We choose to use the binned analysis for gradient measurements since it allows uncertainties for each bin to be calculated and used in fitting of the linear slope. Specifically, the dominant uncertainties including the statistical uncertainty and density shift correction uncertainty are reflected in this analysis.

In the two-region analysis, weighting is done by raw atom counts with no pixel dependent uncertainties. This analysis is meant to demonstrate the differential frequency stability, though as noted in the text the frequency difference between regions is consistent with the redshift reported with the binned analysis.

Comment 12

Fig. 4 It would be interesting to see the QPN limit

We agree with the suggestion and have amended the figure and caption to reflect the predicted QPN limit for the comparison between clock regions.

Comment 13

Methods, in-vacuum cavity: If finesse is ~ 1300 than I would expect a power build up factor at least π times smaller. Does value " ~ 700 " refer to power of incident beam, or (a more conventional) comparison to a standing wave with only a back reflecting mirror? Anyway I do not see how it matches.

Further it is mentioned that cavity mirrors are AR coated @698 nm and that one of the mirrors is rigidly mounted for phase reference for the clock laser. I do not understand how these two facts correspond.

The reference 36 is missing a journal name, and the reference 35 has no information about mirror shielding. And it would be helpful to mention what shielding parameters are used.

The cavity is an over-coupled design meant to optimize the power build-up (circulating intensity/input intensity). We have amended the methods section to clarify this.

“Our over-coupled cavity has a finesse at the lattice wavelength (813 nm) of ~ 1300 and a power buildup factor of ~ 700 (ratio of circulating to input power).”

The bottom mirror of the cavity is rigidly mounted – the clock phase reference is mounted to the same approximate location ensuring the phase of the lattice and clock match. The top cavity mirror has a piezo for cavity frequency stabilization.

We thank the referee for pointing out the missing journal name. References 35 and 36 are included to provide perspective on why the adaption of in-vacuum cavities with moderate finesse within the clock community has been modest despite the clear advantages documented in this paper.

Dimensions for the copper shielding thickness and hole diameter for the lattice beam are now provided in the text.

“Each shield (5 mm thick) has a centered hole of 6 mm diameter to accommodate the optical lattice beam, with shielding performance verified through evaluation of the DC Stark shift systematic.”

Comment 14

Analysis: it is not clear why data with excitation fraction above or below some values should be removed instead of being used for correction.

It would be interesting to have a look at the Gaussian fit of atomic profile at fig 1a, since the data seem to form a plateau in the center. Does cloud position fluctuate at shorter time scale?

We have clarified the text to more clearly motivate these cuts (“Analysis”). They are motivated by avoiding ambiguity and nonlinearity when we perform one-to-one mapping from excitation fraction to frequency of the Rabi lineshape for a range of excitation fractions.

“Frequency corrections based on excitation fraction become ambiguous when the excitation fraction measured is consistent with the Rabi lineshape at multiple detunings. To avoid incorrect frequency corrections, we remove clock locks with excitation fractions above (below) $.903 \times C$ ($.116 \times C$), where C is the Rabi contrast.”

The atom sample shape changes day to day, modestly deviating from a Gaussian. For the systematics and analysis presented in this paper this has no impact since we directly use the raw counts, and as such we have chosen to not include additional fits to atomic distributions.

Within a single day the atomic cloud is stable, especially on short timescales. Analysis of the first hour of data included in Fig. 4 shows that on timescales from 15 to 60 minutes the standard deviation of the center cloud position is ~ 10 microns. We attribute this to a changing magnetic field background impacting the magnetically sensitive

narrow line cooling transition. The magnetic field was recorded during the operation of clock by recording the frequencies of two oppositely Zeeman shifted transitions.

Comment 15

Imaging, first paragraph: does this correspond to geometric calculations (lens magnification and real size of a camera pixel)? Some magnetic field gradient might change the fall acceleration.

The center-of-mass of the atomic sample follows standard kinematic equations of an object in free fall, which when combined with the known gravitational acceleration enable ready calibration of effective pixel size, accounting for camera and imaging system. No geometric calculations are used.

We make no correction for magnetic fields upon the atoms owing to time of flight expansion in the magnetically insensitive ($J=0$) ground state of Sr.

Comment 16

DC Stark: means DC Stark shift gradient?

We have updated the text to reflect that we are discussing a gradient arising from the DC Stark effect. “We evaluate gradients arising from this shift by using in-vacuum quadrant electrodes to apply bias electric fields in all three dimensions.”

Comment 17

Density shift: it seems the minus sign is lost in the last equation.

We quote an absolute value for uncertainties. We have amended the last equation to reflect this, removing the ambiguity of the missing minus sign.